Linear systems and elimination

The exact fifth complete-pivoting growth factor

Difficulty: challenging **Importance:** interesting to specialist

Status: Solution claimed

Rating rationale: Challenging reflects a nonconvex global optimization problem with a high-degree algebraic candidate; specialist impact is an exact low-order pivoting constant. These ratings are historical assessments of the original target.

Full global resolution reported — first public release 12 September 2026

Qianli Ma and Chao Wu, *The exact maximum growth factor for complete pivoting on real 5 × 5 matrices*, report the matching global upper bound $g_5 \le \alpha$, hence $g_5 = \alpha$. The candidate constant and attaining construction remain due to Chen, Edelman and Urschel. The reported upper bound covers all real matrices and every legal complete-pivoting path, including ties and early termination; in particular, it covers this entry's nonsingular real target.

The concise manuscript, Theorem 1.2 (p. 2), states the result; Section 10 (p. 25) composes the global proof. An inverse-elimination representation retains the constraints of the same original matrix. Feasible tail reductions and analytic arguments are combined with exact rational certificates covering the remaining domains. The complete proof supplement and supplementary index supply the detailed arguments and verification map.

Public chronology. The first public source and certificate release, GitHub v1.0.0, was published on **12 September 2026 at 10:37:12 UTC**. The first Zenodo manuscript, v1.0.1, was published on **12 September 2026**. The expanded Zenodo revision, v1.0.2, and GitHub v1.0.2 were published on **30 September 2026**. The concise files included with this catalog contribution reflect the exposition as of **1 November 2026**, rather than a new date for the first public result.

Verification scope. Source code, exact certificates and reproduction instructions are public. The accompanying Lean development is partial: no complete unconditional Lean proof of the target, or Lean verification of the external Python/C++ acceptors, is claimed. This notice reports the authors' complete computer-assisted proof under the catalog's **Solution claimed** evidence level; it does not record an independent verification by this catalog. See the submission and provenance record. The original target and historical supporting results below are retained.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \ldots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1 \le j \le n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Let

$$g_5 = \sup\{\rho_{\mathrm{CP}}(A) : A \in \mathbb{R}^{5\times 5} \text{ nonsingular}\}.$$

Let α be the unique real root in $(4, 5)$ of the integer polynomial P_5 displayed in Equation (2.15) of the primary reference below. This definition specifies an exact algebraic number; numerically $\alpha \approx 4.1325170786324728542$. Prove or disprove

$$g_5 = \alpha.$$

All complete-pivoting paths are included in the supremum. The lower bound $g_5 \ge \alpha$ is already established in the cited paper. The unresolved part is global optimality over all allowable patterns of active constraints, not exact evaluation of a particular numerical example. Order five is independently singled out by the literature as the first unresolved dimension; this catalog does not split the same question into entries for every subsequent order.

Reference

Chen, Edelman, and Urschel, *The largest 5th pivot may be the root of a 61st degree polynomial*, February 2026, Equation (2.15), Conjecture 2.1, Theorem 2.2, and Theorem 3.5. The rigorous interval established in that reference is $\alpha \leq g_5 \leq 4.84$.

Earlier status check — 2026-09-08

Searches for the paper title and *complete pivoting maximum five 2026* found no global proof. Shah–Urschel's August asymptotic result leaves this exact finite-dimensional optimum undetermined.

Audit update — 2026-09-10

The February 2026 primary preprint still presents the degree-61 candidate as conjecturally optimal, rather than proving the matching upper bound. Title-based and fifth-pivot resolution searches found no later exact determination. The broader complete-pivoting bounds do not certify this finite-order optimum.

Reviewed supporting results — 13 September 2026

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The submitted report, Section 5 (Theorem 5.1), certifies strict local maximality of the algebraic candidate in all 24 free entries with the normalization and diagonal complete-pivoting path fixed. Sections 3–4 reconstruct the candidate and degree-61 polynomial; Section 6 provides a strictly feasible rational witness. The candidate and established lower bound remain attributed to Chen, Edelman and Urschel. Section 7 certifies an auxiliary relaxation optimum of 81/16, which is not a feasible complete-pivoting growth example.

The independent informal AI-agent review supports the explicitly limited claims. **Status remained Open at that review:** the work supplies supporting local and certificate results, but no matching global upper bound and no counterexample. All original matrices and admissible pivot paths remain part of the target.

Submission record and verified affiliation. Substantial AI assistance is disclosed. The review is informal automated review, not external human peer review or formal verification. No Lean verification was performed.